

nlp-practical-5

April 7, 2026

#23CO315 - Piyusha Supe **NLP Practical 5**

Morphology is the study of the way words are built up from smaller meaning bearing units. Study and understand the concepts of morphology by the use of add delete table

```
[2]: import pandas as pd

# Sample words
words = [
    "playing", "played", "plays",
    "running", "runner",
    "happiness", "unhappy",
    "cats", "boxes"
]

# Common suffixes
suffixes = ["ing", "ed", "s", "er", "ness"]

results = []

for word in words:
    base = word

    for suf in suffixes:
        if word.endswith(suf):
            base = word[:-len(suf)] # delete suffix

            results.append({
                "Original Word": word,
                "Operation": "DELETE",
                "Removed": suf,
                "Base Form": base
            })

    # Add back transformation
    results.append({
        "Original Word": base,
        "Operation": "ADD",
```

```

        "Added": suf,
        "New Word": base + suf
    })

# Convert to DataFrame
df = pd.DataFrame(results)

df

```

```

[2]:
Original Word Operation Removed Base Form Added New Word
0      playing  DELETE      ing    play  NaN      NaN
1        play    ADD      NaN    NaN  ing    playing
2      played  DELETE      ed    play  NaN      NaN
3        play    ADD      NaN    NaN  ed    played
4       plays  DELETE      s    play  NaN      NaN
5        play    ADD      NaN    NaN  s    plays
6    running  DELETE      ing   runn  NaN      NaN
7       runn    ADD      NaN    NaN  ing    running
8     runner  DELETE      er   runn  NaN      NaN
9       runn    ADD      NaN    NaN  er    runner
10  happiness  DELETE      s  happines  NaN      NaN
11   happines    ADD      NaN    NaN  s    happiness
12  happiness  DELETE  ness   happi  NaN      NaN
13    happi    ADD      NaN    NaN  ness    happiness
14     cats  DELETE      s    cat  NaN      NaN
15     cat    ADD      NaN    NaN  s    cats
16   boxes  DELETE      s   boxe  NaN      NaN
17    boxe    ADD      NaN    NaN  s    boxes

```

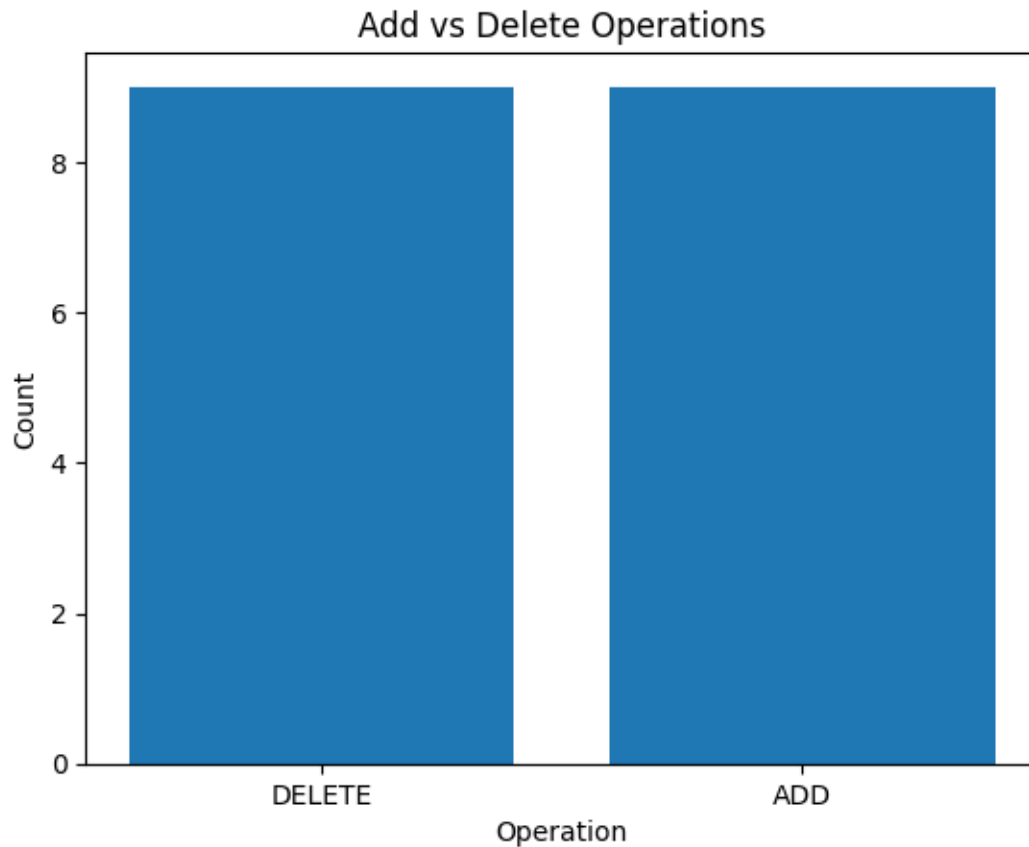
```

[3]: import matplotlib.pyplot as plt

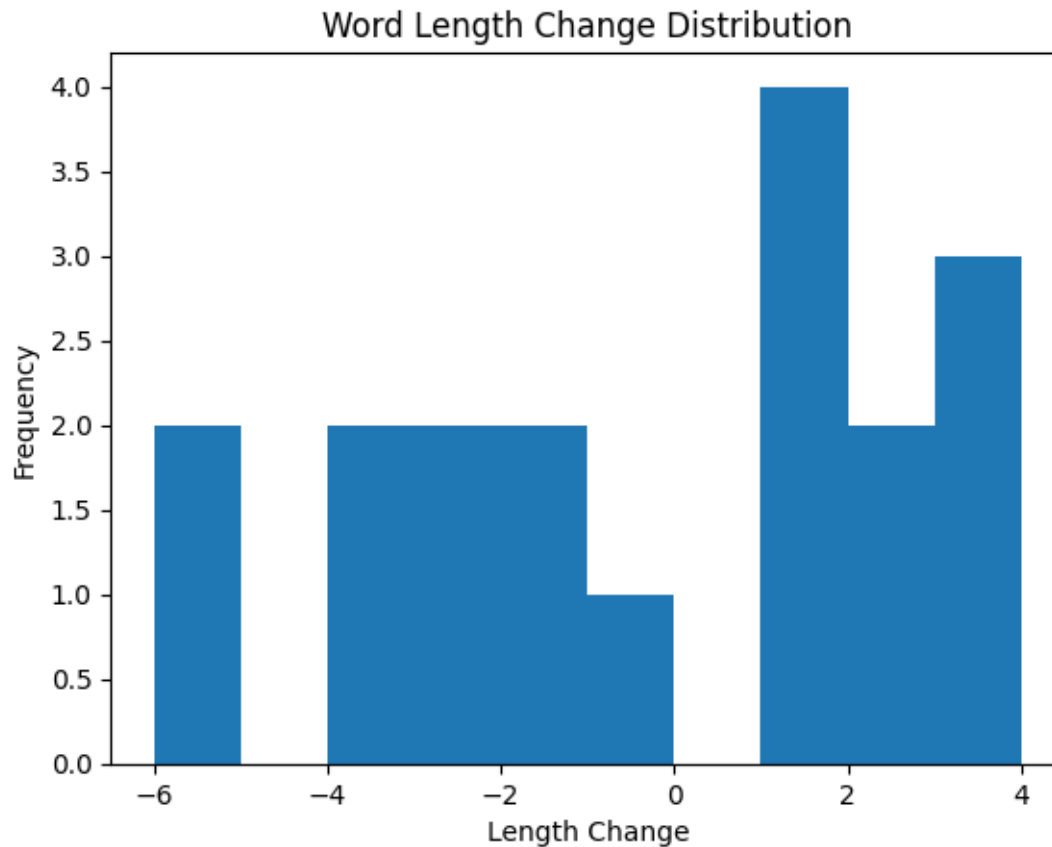
counts = df['Operation'].value_counts()

plt.figure()
plt.bar(counts.index, counts.values)
plt.title("Add vs Delete Operations")
plt.xlabel("Operation")
plt.ylabel("Count")
plt.show()

```



```
[4]: df['Length Change'] = df.apply(  
    lambda x: len(str(x.get("New Word", x.get("Base Form")))) - len(x['Original_  
    ↪Word']),  
    axis=1  
)  
  
plt.figure()  
plt.hist(df['Length Change'], bins=10)  
plt.title("Word Length Change Distribution")  
plt.xlabel("Length Change")  
plt.ylabel("Frequency")  
plt.show()
```



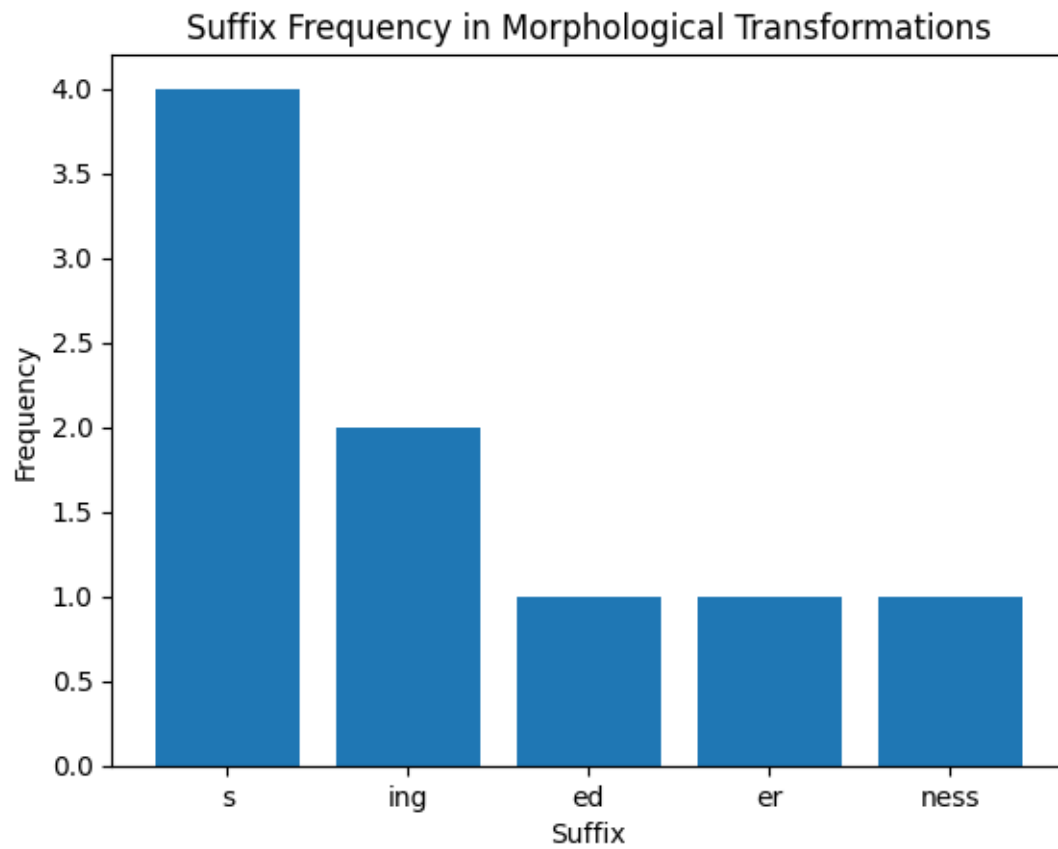
```
[5]: # Extract suffixes used
suffix_used = []

for _, row in df.iterrows():
    if row['Operation'] == 'DELETE':
        suffix_used.append(row['Removed'])

# Count frequency
suffix_counts = pd.Series(suffix_used).value_counts()

# Plot
import matplotlib.pyplot as plt

plt.figure()
plt.bar(suffix_counts.index, suffix_counts.values)
plt.title("Suffix Frequency in Morphological Transformations")
plt.xlabel("Suffix")
plt.ylabel("Frequency")
plt.show()
```



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